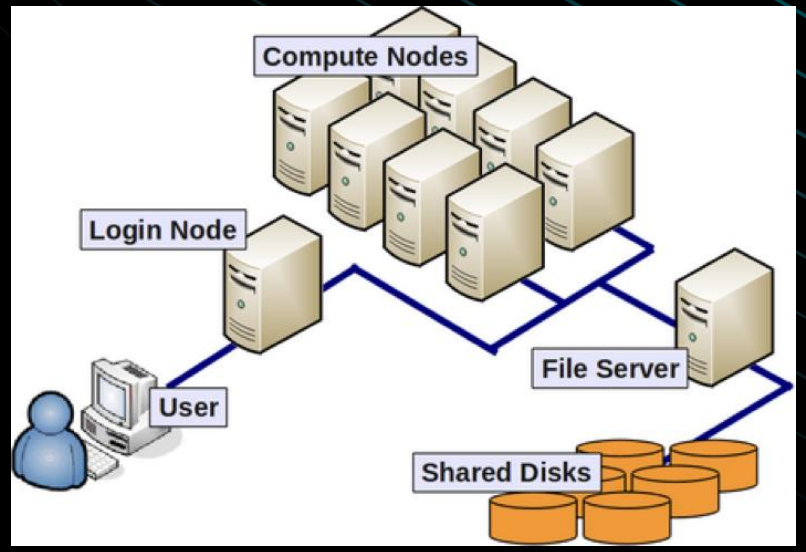
HCEMM
Translational Medicine

HUNGARIAN NATIONAL LABORATORY

Cutting-Edge Bioinformatics in a High-Performance Computing Environment



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Workshop overview

- 1st day – Introduction to Linux and the Shell environment
  - The Linux filesystem, nice command-line utilities, remote access to a Linux server
- 2nd day - Running bioinformatics tools for data processing and analysis
  - Step by step, from raw sequencing data to differential analysis of gene expression between two health conditions
- 3rd day – Using a workload manager to handle large-scale datasets
  - How to submit data analyses to Slurm
- 4th day – Results interpretation, Q & A, What comes next
  - Convert experimental results into translational results

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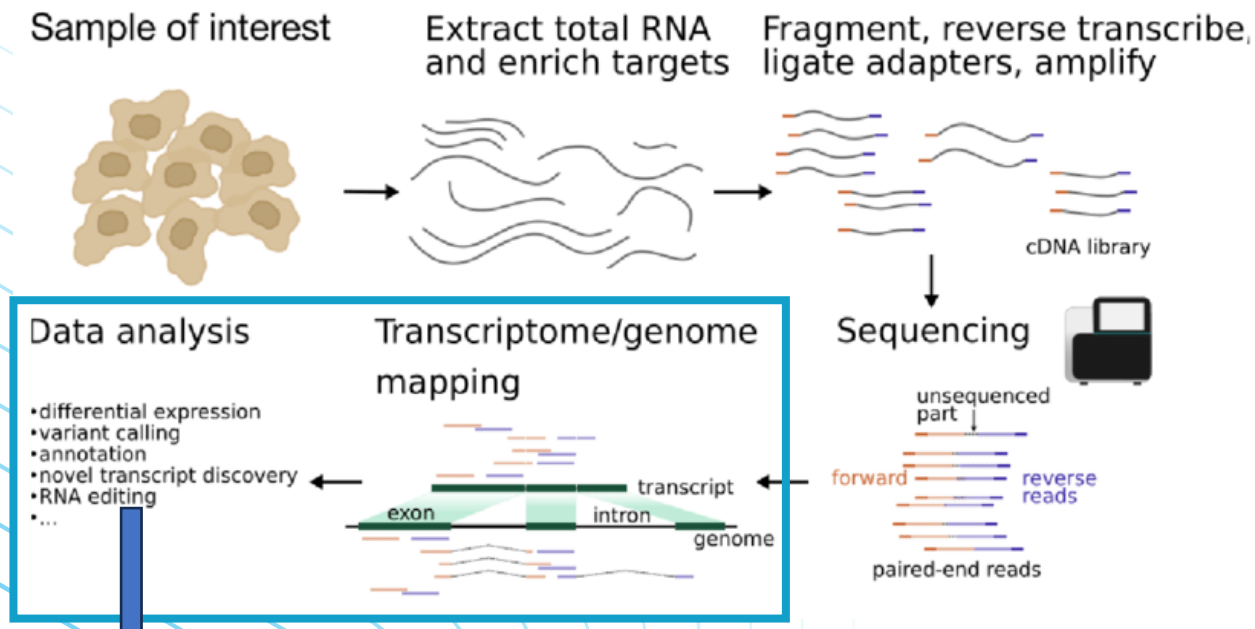
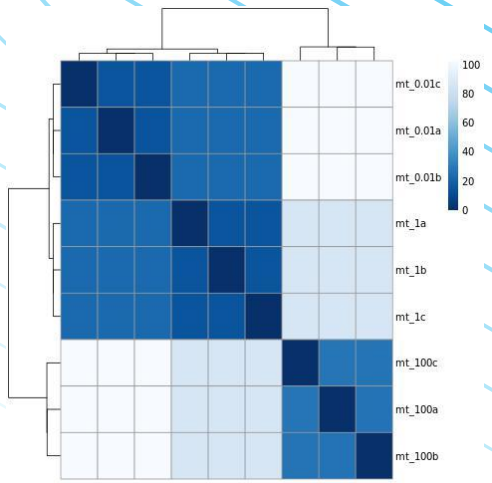
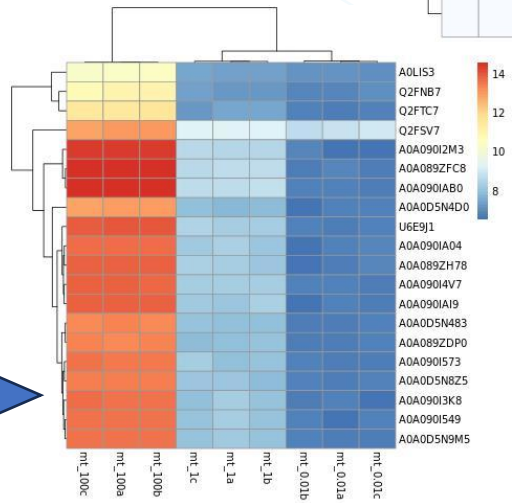
Quick overview of the bioinformatics pipeline

We will be running, step by step, a pipeline for estimating gene expression, and statistically identify differential expression of genes between two medical conditions.



Bioinformatics data analysis





Bioinformatics results, and their translational into medicine

Image from: https://doi.org/10.7287/peerj.preprints.27283

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Before the workshop – FileZilla installation

- Go to: https://filezilla-project.org/
- Select your OS -> “Download FileZilla Client”.
- Or just chose from the list here:
  - Windows 64 bits (most of current Windows)
  - Windows 32 bits
  - MacOS ARM64 (more modern Macs)
  - MacOSX
  - Linux 64 bits (most Linuxes)
  - Linux 32 bits
- And follow the instructions of the installer

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Before the workshop – testing access to Puli

- Go to: https://puli.hcemm.eu/
- Input the credentials provided by email.
- If they work, you are all set!
- Otherwise, please email us:
  - maria.kavoosi@hcemm.eu
  - joao.sequeira@hcemm.eu

See you at the workshop!

And for any questions, don’t hesitate contacting:

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